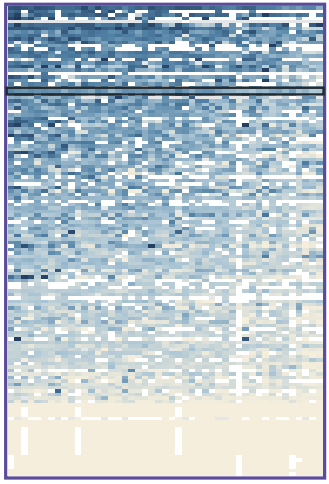
Is this reaction worth modelling for substrate pairs not yet run?

19 arrays · 17 reaction families · 378,070 yields

One screening table

Grignard addition: 136 pairs × 47 conditions



rows: pairs · columns: conditions

Read before modelling

between-pair share 71%
pairing sparsity 0.55

both track the loss: 0.67, 0.64
(rank correlations, 11 arrays)

Held-out wells



a few wells are hidden; the pair's other wells stay in training: the model recalls

random splits: Ahneman 2018, Schwaller 2021

New pairs



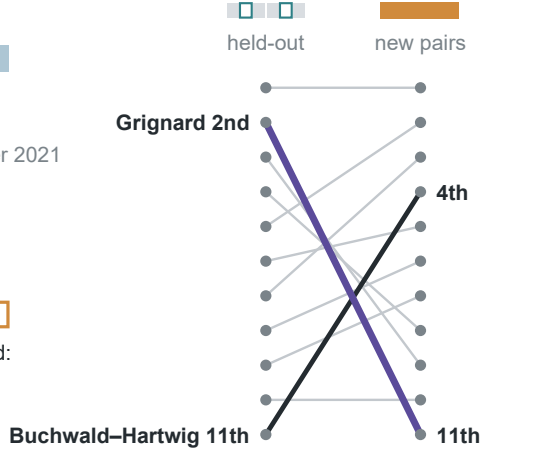
the whole pair and its plates are withheld: the model must predict the pair

withheld components or pairs: Sandfort 2020, Ismail 2026, Götz 2025, Neves 2026

pair mean R^2 0.83 → 0.23
within pair 0.34 → 0.33
median R^2, 11 arrays

Does the order of reactions survive?

not compared before: this work



rank correlation 0.02

11 arrays averaging ≥5% yield

Grignard addition

